MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3740 SECONDARY CYANIDE DESTRUCTION
DESIGN OF TANK FOUNDATION

Document Number: **MD-216-3740-EG-CV-CAL-1001**
Revision: **D01**
Revision Date: **28.11.2023**
Status: **AS BUILT**

Outotec Document Number:	900707	GAU01	RN14	CED01	37402	OU100705878	2
L&T Document Number:	O19045	450	C	K	24537	001	2

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	27.10.2020	FIRST ISSUE	SOR	SJM	AG	
C01	18.11.2020	ISSUED FOR CONSTRUCTION	SOR	SJM	AG	
D01	28.11.2023	AS BUILT	SOR	SJM	AG	

TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS 3

1. INTENT 4
2. EXECUTIVE SUMMARY 4

3. SCOPE 4

4. DESIGN REQUIREMENTS 5

4.1 Design Loads 5

5. CONCRETE FOUNDATIONS 5

5.1 Design 5

5.2 Materials 5

5.3 Foundations 5

5.4 Minimum Concrete Cover 6

5.5 Allowable Soil Bearing Capacity 6

6. DESIGN CALCULATION FOR FOUNDATION 7

References 12

ATTACHMENT 1 LOAD DATA DRAWINGS 13

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

1. INTENT

This document is to prepare design of foundation for **Secondary Cyanide Destruction Reactor (3740-REA-001)**. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel. For member strength calculation force values are directly taken from Equipment load data drawings & designed in excel sheet.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Load Data drawings attached in Appendix A. Also referred Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

5.3 Foundations

Foundation shall be Earth Foundation with a Concrete Ring wall for Tanks. The design calculations are prepared as per recommendation of standard API 650 - Annex B.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

5.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation	50
Exposed to weather or contact with ground	Wall	50

5.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 450 Kn/m^2 at 1.20 m below EGL for the tank. (Refer Table 5 - BH-27 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

6. DESIGN CALCULATION FOR FOUNDATION

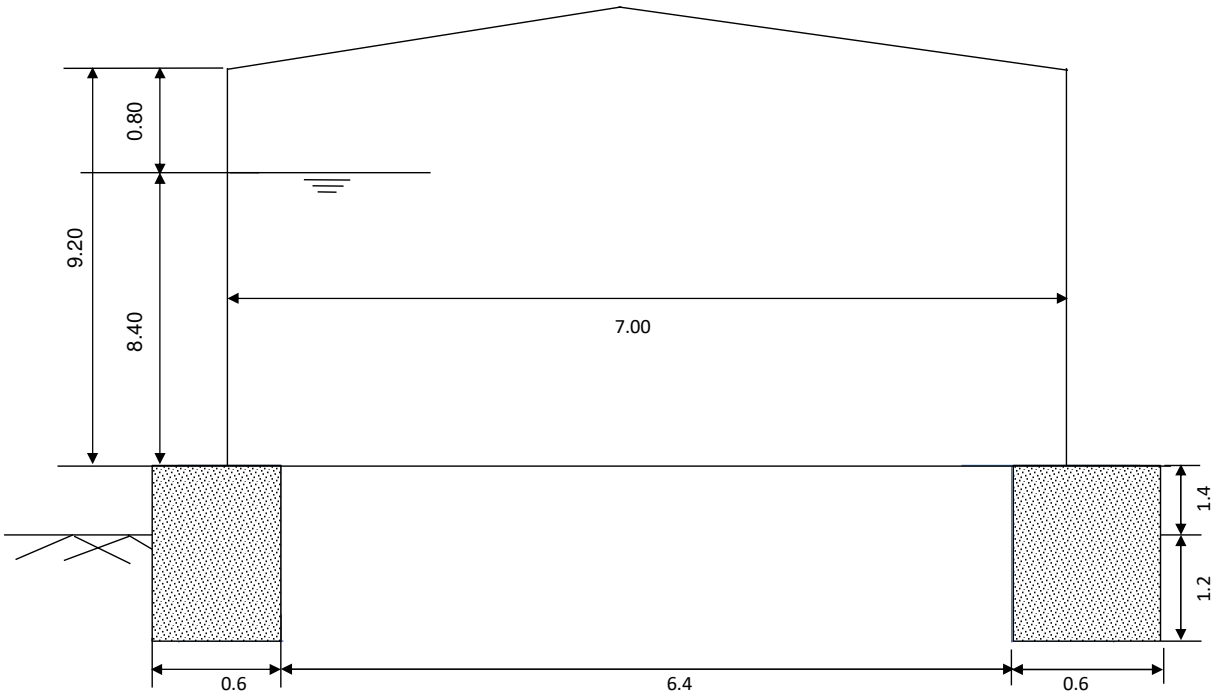
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Tank Loading Data for 3740-REA-001

	Loading Type		
Weight, V	Empty (W)	261.5	kN
	Operating (Wo)	3233	kN
	Test (Wt)	3541	kN

Summary of Tank Specification



Diameter of Tank, D	=	7.0 m
Height of Tank, H	=	9.2 m
Height of Tank Fluid, H_L	=	8.4 m
Inner Diameter of Concrete Ring Wall, ID	=	6.4 m
Outer Diameter of Concrete Ring Wall, OD	=	7.6 m
Height of Concrete Ring Wall, h	=	2.60 m
Thickness of Concrete Ring Wall, t	=	0.6 m
Top of Concrete above Finished Ground Level	=	1.4 m
Unit Weight of Liquid, γ_L	=	11.0 kN/m^3
Unit Weight of Soil, γ_s	=	20.0 kN/m^3
Unit Weight of Concrete, γ_c	=	24.0 kN/m^3

Wind Load & Seismic Load

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)
Wind Loads	Empty	62.00	334.80
	Operating	62.00	334.80
Seismic Loads	Operating	148.00	695.00

Foundation Weight

a) Ring Wall Weight = 823.35 kN

a) Inside Soil Weight = 1672.84 kN

Sectional Area

Net cross-sectional area for empty condition, Ae = 13.19 m^2

Full cross-sectional area for operating condition, Ao= 45.36 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze = 21.42 m^3

Sectional modulus for operating condition,Zo = 43.10 m^3

Check of Minimum Wall Thickness (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to tank,minimum thickness of concrete wall required to equalize soil pressures between under tank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma c\ h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + \gamma_L H_L / 2 + \gamma c\ h$

or, $t = 2w / (\gamma_L * H_L - 2h(\gamma_c - \gamma s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D)$ = 11.89 kN/m

So, Required Min Wall THK.= 0.3322 m = 332.2 mm < 600 mm OK

Basic Load Cases

1. Tank Dead Load (D)
2. Operational Fluid Load (F1)
3. Test/Max Fluid Load (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301,2007 , Cl No. 2.4.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101	D + F1	3233.00	0.00	0.00
Short term	Empty	LC102	0.6 D + W	156.90	62.00	334.80
	Operating	LC103	D + F1 + W	3233.00	62.00	334.80
		LC104	D + F1+ 0.7 E	3233.00	103.60	486.50
		LC105	0.6 (D + F1) + 0.7 E	1939.80	103.60	486.50
	Test	LC106	D + F2	3541.00	0.00	0.00

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 450 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 598.5 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Qmax= V/A - M/Z kN/m^2	Qall/ Qmax	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	5729.18	0.00	126.29	126.29	3.56	OK
LC102	980.25	496.00	97.44	51.14	6.14	OK
LC103	5729.18	496.00	137.80	114.78	4.34	OK
LC104	5729.18	755.86	143.83	108.75	4.16	OK
LC105	4435.98	755.86	115.32	80.25	5.19	OK
LC106	6037.18	0.00	133.08	133.08	3.38	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of foundation level	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
	V (kN)					
LC101	5729.18	0.4	2291.67	0.00	-	-
LC102	980.25	0.4	392.10	62.00	6.3	OK
LC103	5729.18	0.4	2291.67	62.00	37.0	OK
LC104	5729.18	0.4	2291.67	103.60	22.1	OK
LC105	4435.98	0.4	1774.39	103.60	17.1	OK
LC106	6037.18	0.4	2414.87	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of foundation level	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
	V (kN)					
LC101	5729.18	3.8	21770.90	0.00	-	-
LC102	980.25	3.8	3724.94	496.00	7.5	OK
LC103	5729.18	3.8	21770.90	496.00	43.9	OK
LC104	5729.18	3.8	21770.90	755.86	28.8	OK
LC105	4435.98	3.8	16856.74	755.86	22.3	OK
LC106	6037.18	3.8	22941.30	0.00	-	-

Ultimate Strength Design

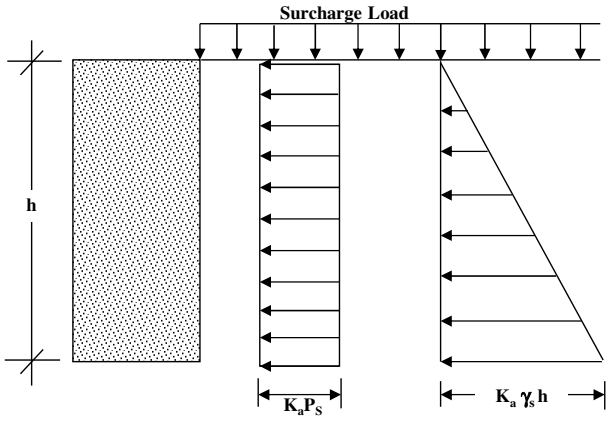
Load Combination (As per SBC-304 Table 5.3.1)

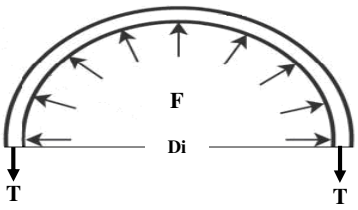
Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	4526.20	0.00	38.48	33.67
Short term	Empty	LC102	0.9 D + 1.6 W	235.35	535.68	38.48	33.67
	Operating	LC103	1.2 (D + F1) + 1.6 W	3879.60	535.68	38.48	33.67
		LC104	1.2 (D + F1) + E	3879.60	695.00	38.48	33.67
		LC105	0.9 (D + F1) + E	2909.70	695.00	38.48	33.67
	Test	LC106	1.4 (D + F2)	4957.40	0.00	38.48	33.67

Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m^2
LC101	4526.20	0.00	117.61
LC102	235.35	535.68	22.02
LC103	3879.60	535.68	116.72
LC104	3879.60	695.00	121.45
LC105	2909.70	695.00	96.25
LC106	4957.40	0.00	128.82

Maximum Surcharge Load, Ps = 128.82 kN/m^2





Where, K_a = coefficient of earth pressure at rest
= 0.5

P1 = $K_a * P_s$ = 64.41 KN/m^2

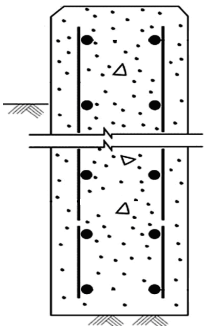
P2 = $1.4 * \gamma_S * h * K_a$ = 36.40 KN/m^2 for LC106 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 214.78 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 687.296 KN



Hoop Tension Bar

1. Horizontal Reinforcement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = T/(φ x Fy) = 3900 mm^2

Minimum As = 0.0025 x b x D = 3900 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 24 Nos. D16= 4355.001 mm^2 OK

2. Vertical Reinforcement

Minimum As = 0.0015 * b * D = 900 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row * 6 each D10 (@ 167mm)

=942mm2 > Required As OK

REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

PROPRIETARY & CONFIDENTIAL

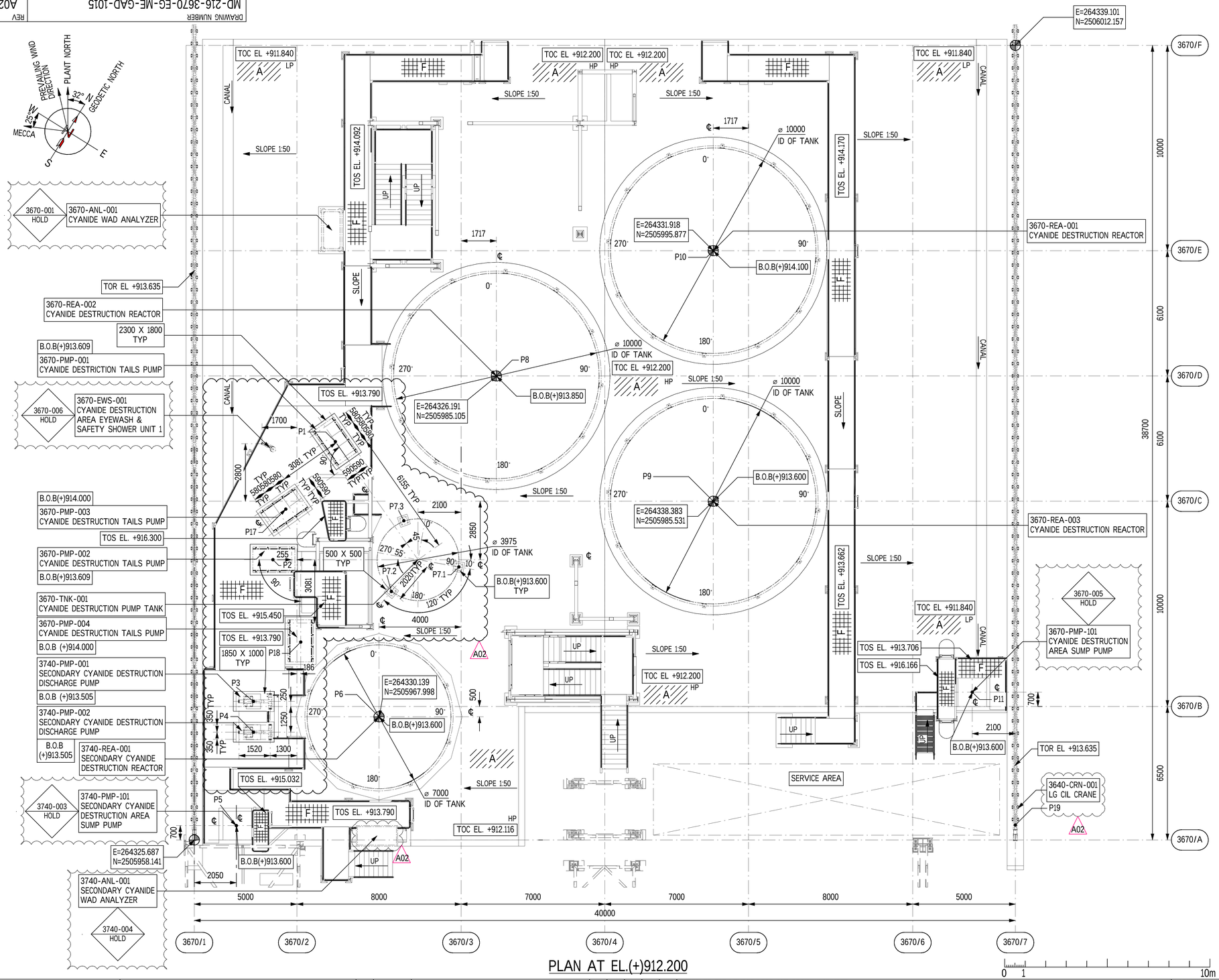
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ATTACHMENT 1 LOAD DATA DRAWINGS

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DRAWING NUMBER MD-216-3670-EG-ME-GAD-1015 REV A02



PLAN AT EL.(+)912.200

0 1 10m

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ALKALI / ACID RESISTANT COATING
- CHECKERED PLATE
- GRATING
- GRATING - FRP
- ACID RESISTANT BRICKLINING
- BRICK WALL
- CONCRETE WALL

GENERAL NOTES, LEGEND

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
T.O.R. = TOP OF RAIL
LP = LOW POINT
HP = HIGHT POINT

DIRECTION OF FORCES AND MOMENTS

06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
OPERATING PLATFORM-GRATING 5 kN/m² GROUNDFLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²
HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8 kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1,6 FOR STRENGTH DESIGN.

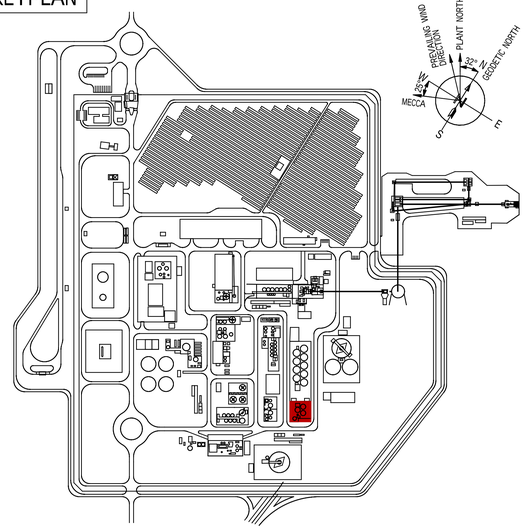
13 - REPORTED SEISMIC LOAD ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301-2007. HOWEVER, IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNER.

14 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)912.200 m.

15 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

16 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3000-EG-GM-LST-1010	HOLD LIST								
MD-216-3670-EG-ME-GAD-1010	GENERAL ARRANGEMENT DRAWING, PLAN AT EL. +912.200								
MD-216-3670-EG-ME-GAD-1017	LOAD TABLE	A02	24-07-2020	ISSUED FOR DESIGN	MM	TV	SC	RL	
MD-216-3670-EG-ME-GAD-1016	LOAD PLAN, PLAN VIEW	A01	31-01-2020	ISSUED FOR APPROVAL	BR	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER 019045_450_M_D_03258_001 — REVISION 1

OUTOTEC DRAWING NUMBER 900707 _ GAU01 _ RN12 _ MLD14 _ 36701 — DOCUMENT ID OU100667835 — REVISION 1

LARSEN & TOUBRO

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it or be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن Saudi Arabian Mining Company شركة المعادن العربية السعودية

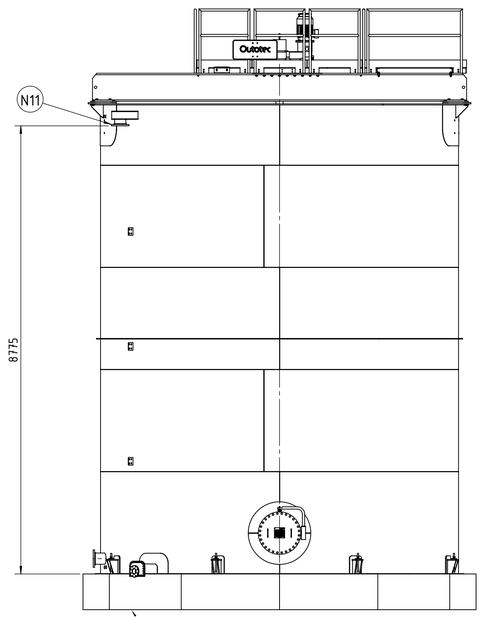
This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

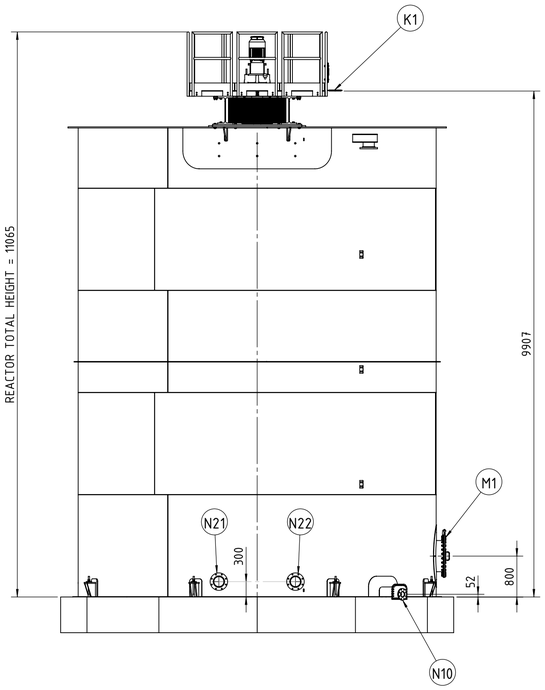
DRAWN	BR	31-01-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	31-01-2020	LOAD PLAN
ENGINEER	SC	31-01-2020	PRIMARY CYANIDE DESTRUCTION
APPROVED	RL	31-01-2020	PLAN AT EL. +912.200
EMPLOYER			
SCALE 1:100		A1	DRAWING NUMBER MD-216-3670-EG-ME-GAD-1015 — REV A02

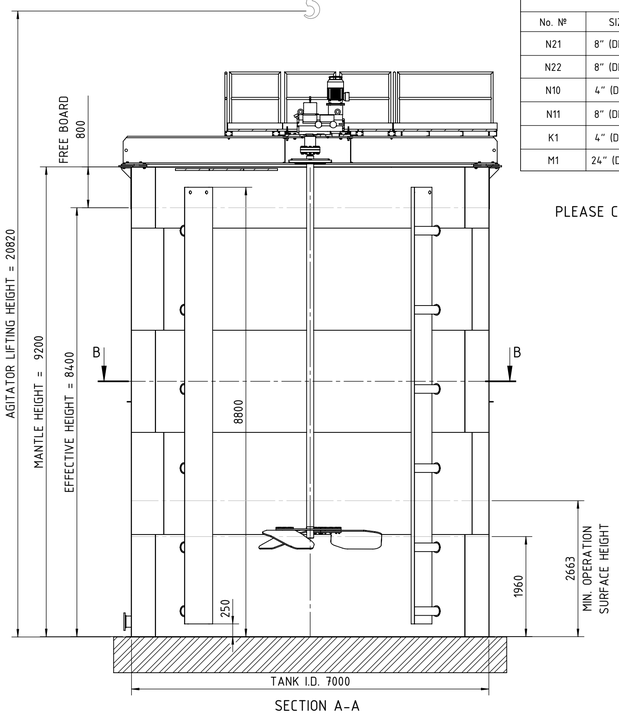
DRAWING NUMBER MD-216-3740-EG-ME-GAD-1001
REV D01

NOZZLE TABLE								
No. №	SIZE	RATING	DIRECTION	HEIGHT	STANDARD	SERVICE	QUANTITY	REMARKS
N21	8" (DN200)	CLASS 150	270°	300	ASME B16.5	SLURRY OUTLET	1	
N22	8" (DN200)	CLASS 150	270°	300	ASME B16.5	SLURRY OUTLET	1	
N10	4" (DN100)	CLASS 150	225°	52	ASME B16.5	DRAIN	1	
N11	8" (DN200)	CLASS 150	235°	8775	OUTOTEC	OVERFLOW	1	
K1	4" (DN100)	CLASS 150	180°	9907	ASME B16.5	LEVEL MEASUREMENT	1	
M1	24" (DN600)	-	180°	800	OUTOTEC	MANHOLE	1	

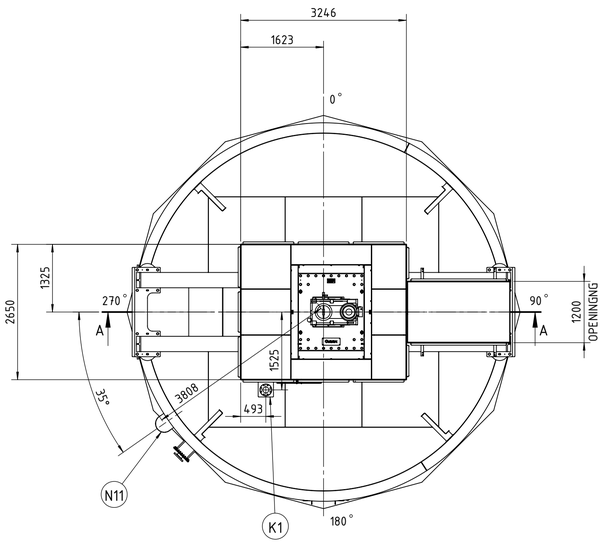
PLEASE CHECK DATASHEET MD-216-3380-EG-ME-DAT-1002 FOR PROCESS VALUES

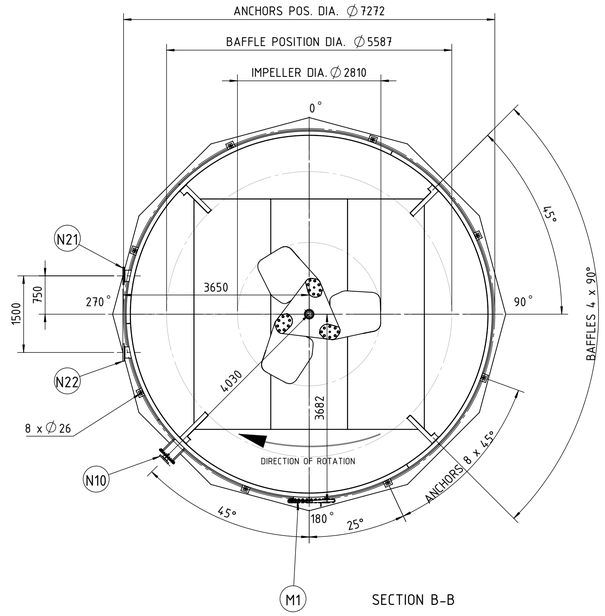




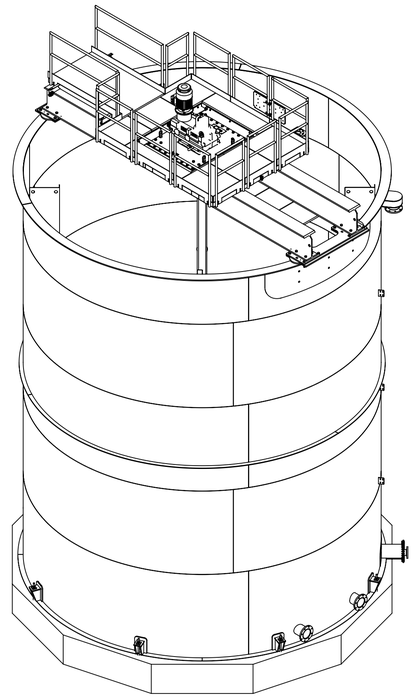


SECTION A-A





SECTION B-B



Secondary cyanide destruction reactor	
AGITATOR WEIGHT APPR. (kg)	8716
DEAD WEIGHT (kg)	13993
OPERATIONAL WEIGHT (kg)	346009
MAXIMUM WEIGHT (kg)	376809

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3740-EG-ME-PLN-1001	FOUNDATION AND LOAD PLAN								
MD-216-3740-EG-ME-DAT-1002	EQUIPMENT DATASHEET	D01	28.11.2023	AS BUILT	MH	TM	JR	GR	
MD-216-3740-EG-PR-PID-1001	P&I DIAGRAM	C01	19.05.2020	APPROVED FOR CONSTRUCTION	MH	TM	JR	GR	
MD-216-3740-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	14.01.2020	ISSUED FOR APPROVAL	MH	TM	JR	GR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

This drawing is the property of *LARSEN & TOUBRO* and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-G-02923-005 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707 GAU01 RN14 MTL03 37401 — DOCUMENT ID: OU601239225 — REVISION: 2

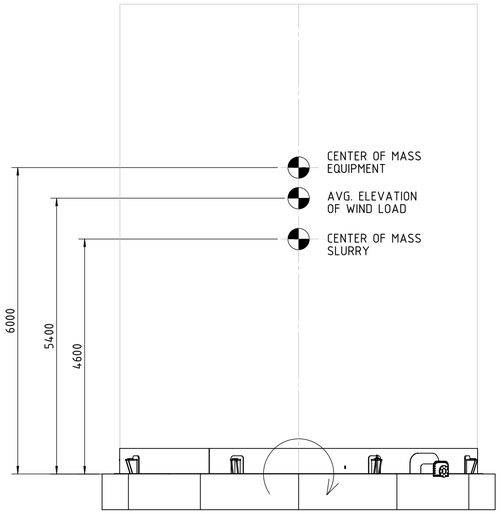
DRAWN	mikhov	15.05.2020
CHECKED	tuoman	19.05.2020
ENGINEER	juhris	19.05.2020
APPROVED	GeoRie	19.05.2020
EMPLOYER		
SCALE	1:60	A1

MANSOURAH-MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
SECONDARY CYANIDE DETOXIFICATION
3740-REA-001 SECONDARY CYANIDE DESTRUCTION REACTOR

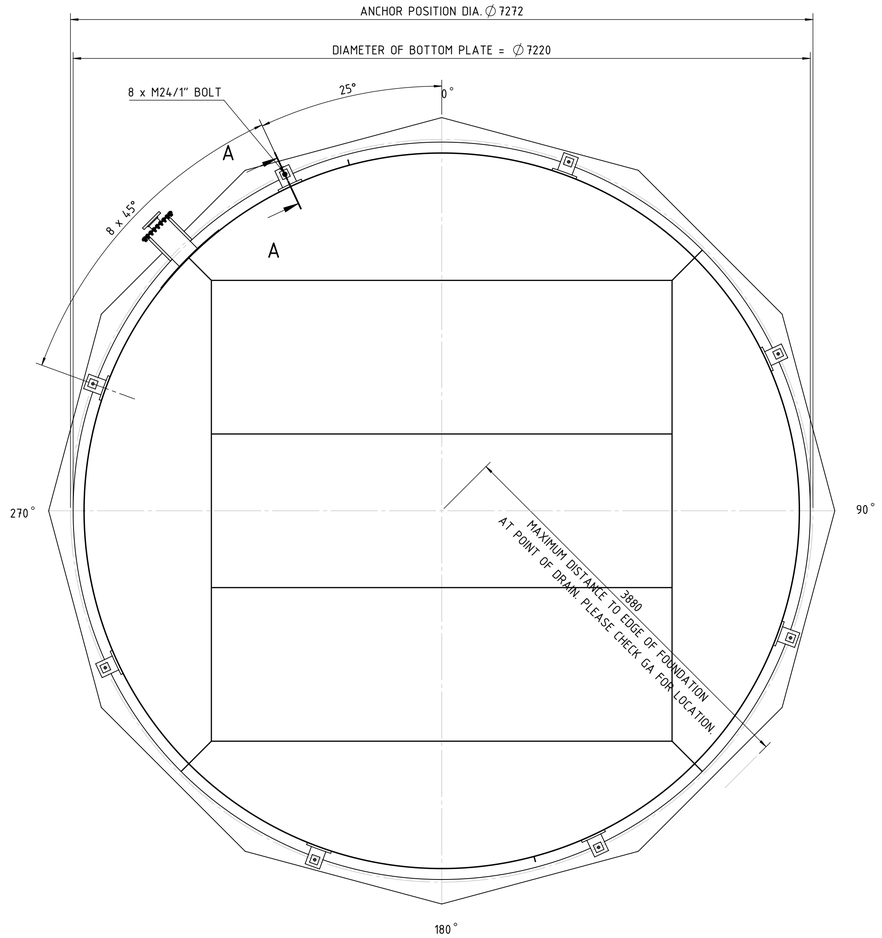
DRAWING NUMBER: MD-216-3740-EG-ME-GAD-1001 — REV: D01

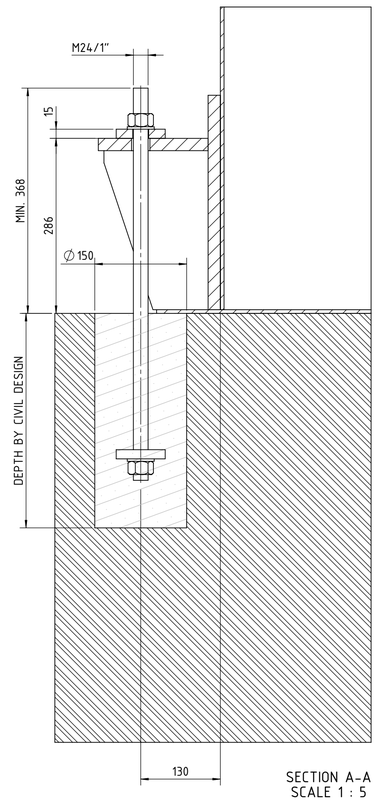
DRAWING NUMBER	REV
MD-216-3740-EG-ME-PLN-1001	D01

LOADING NOTES:	
1.	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2.	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
3.	REPORTED SEISMIC LOADS ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC 301 - 2007. HOWEVER IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNERACCORDING SBC 301 - 2007.
4.	REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC 301 - 2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGTH DESIGN
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL
CONCRETE NOTES: (WHERE SHELLS ARE INSTALLED ON CONCRETE AND HOLD DOWN BOLTS ARE USED)	
1.	CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT



SCALE 1 : 60





SECTION A-A
SCALE 1 : 5

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0.02g
-Ss=0,10g
-max. ground elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

SECONDARY CYANIDE DESTRUCTION TANK	
DEAD WEIGHT OF EQUIPMENT	23950 kg
OPERATIONAL WEIGHT OF SLURRY	323300 kg
MAXIMUM WEIGHT OF SLURRY	354100 kg
EXTERNAL PLATFORM LOAD	2200 kg
NORMAL OPERATING SPEED	28,5 rpm
WIND LOAD	62 kN
SEISMIC LOAD	148 kN
SEISMIC TORQUE AT FOUNDATION LEVEL	695 kNm

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3740-EG-ME-GAD-1001	GENERAL ARRANGEMENT DRAWING								
MD-216-3740-EG-ME-DAT-1002	EQUIPMENT DATASHEET	D01	28.11.2023	AS BUILT	VP	TM	JR	GR	
MD-216-3740-EG-PR-PID-1001	P&I DIAGRAM	C01	03.03.2020	APPROVED FOR CONSTRUCTION	VP	TM	JR	GR	
MD-216-3740-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	14.01.2020	ISSUED FOR APPROVAL	JJ	TM	JR	GR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

This drawing is the property of *LARSEN & TOUBRO* and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

LARSEN & TOUBRO DRAWING NUMBER		REVISION
019045-450-M-D-02924-004		3
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707 GAU01 RN14 MLD15 37401	OU601269174	3

DRAWN	vilpoy	28.02.2020
CHECKED	tuoman	03.03.2020
ENGINEER	juhris	03.03.2020
APPROVED	GeoRie	03.03.2020
EMPLOYER		
SCALE	1:30	A1

MANSOURAH - MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
SECONDARY CYANIDE DETOXIFICATION
3740-REA-001 SECONDARY CYANIDE DESTRUCTION REACTOR

DRAWING NUMBER	REV
MD-216-3740-EG-ME-PLN-1001	D01